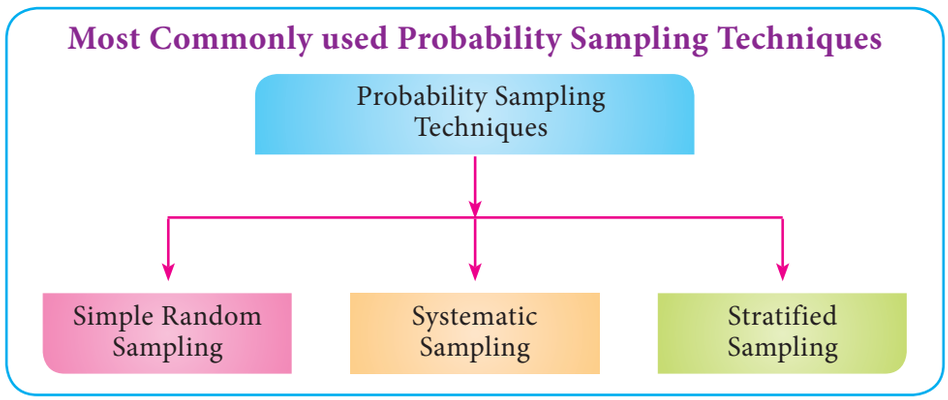
2.7 Probability sampling:

Probability sampling or random sampling involves a random selection process to include a unit of the population in the sample. In random sampling, all items have a positive probability of selection. Random sampling techniques remove the personal bias of the investigators. Some of the important methods of probability sampling are: Simple Random Sampling, Stratified Random Sampling and Systematic Sampling.



2.7.1 Simple random sampling

This method selects a sample in such a way that each possible sample to have an equal probability of being selected or each item in the entire population to have an equal probability of being included in the sample. The following are the methods of selecting a simple random sample.

(1) Lottery Method

Suppose that we have to select a random sample of size n from a finite population of size *N*. First assign numbers 1 to N to all the N units of the population. Then write numbers 1 to *N* on different identical slips or cards so that a card is not distinguishable from another. They are folded and mixed up in a drum or a box or a container. A blindfold selection is made. The selected card may be replaced or may not be replaced before the next draw. The Required numbers of slips are selected for the desired sample size under any one of these methods forms a simple random sample. The selection of items thus depends on chance.

Simple random sampling without replacement (SRSWOR):

If the selected cards are not replaced before the next draw, such a sampling is called without replacement.

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